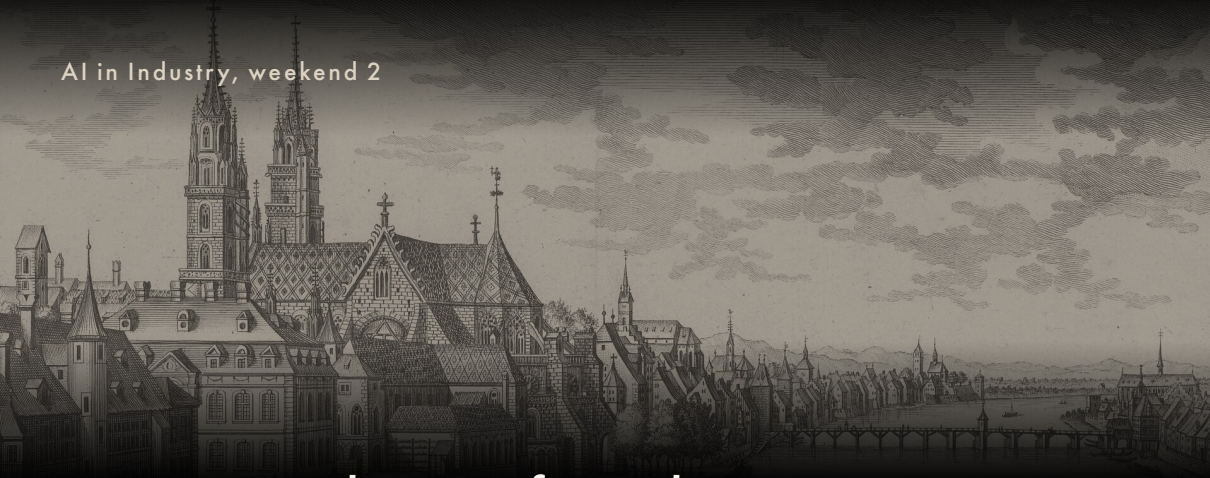
AI in Industry, weekend 2



Estimating chances from the past

A book from Basel, 1713

Emanuel Büchel, *Prospect des Grossen Münsters...*, 1761, detail.
Zentralbibliothek Zürich. Wikimedia Commons, public domain



A professor in Basel

Switzerland, around 1690

Jakob Bernoulli teaches mathematics in Basel. The chances of a die can be counted before the throw: one face in six. But nobody can count in advance the chance that a man dies within ten years, or that it rains tomorrow.

Emanuel Büchel, *Prospect des Grossen Münsters...*, 1761, detail. Zentralbibliothek Zürich. Wikimedia Commons, public domain



A professor in Basel

Switzerland, around 1690

Jakob Bernoulli teaches mathematics in Basel. The chances of a die can be counted before the throw: one face in six. But nobody can count in advance the chance that a man dies within ten years, or that it rains tomorrow.

How could anyone know such a chance?

Emanuel Büchel, *Prospect des Grossen Münsters...*, 1761, detail. Zentralbibliothek Zürich. Wikimedia Commons, public domain



Quod a priori elicere non datur, saltem a posteriori … eruere licebit.

What we cannot find out beforehand, we may at least find out afterwards.

Jakob Bernoulli, *Ars Conjectandi*, Basel, 1713, part IV, chapter 4.
English: lecturer

Emanuel Büchel, *Prospect des Grossen Münsters…*, 1761, detail.
Zentralbibliothek Zürich. Wikimedia Commons, public domain



Niklaus Bernoulli, portrait of Jakob Bernoulli, 1687, detail. Alte Aula of the University of Basel. Wikimedia Commons, public domain

Jakob Bernoulli

1655 to 1705

"Suppose that of three hundred men of the age and constitution of Titius, two hundred died within ten years. Then you can conclude, safely enough, that there are twice as many cases in which Titius pays his debt to nature within ten years as cases in which he lives longer."

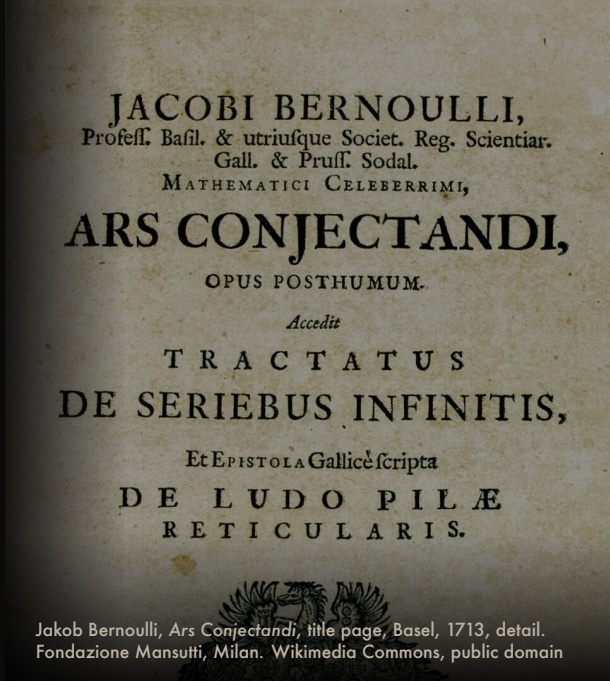
Jakob Bernoulli, *Ars Conjectandi*, Basel, 1713, part IV, chapter 4. English: lecturer

Ars Conjectandi

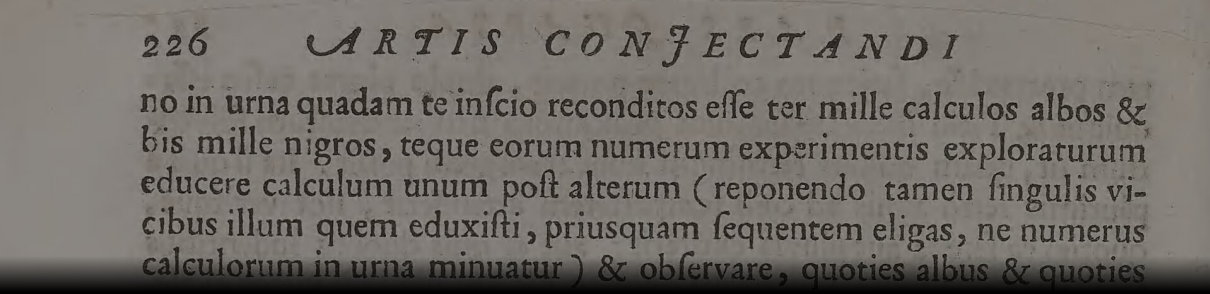
The Art of Conjecturing, 1713

"One or two experiments are not enough; a great number is needed. Even the most stupid person, by some instinct of nature, on his own and without any instruction (which is truly astonishing), knows that the more observations are taken, the smaller the danger of missing the mark."

Jakob Bernoulli, *Ars Conjectandi*, Basel, 1713, part IV, chapter 4. English: lecturer


JACOBI BERNOULLI,
Profess. Basil. & utriusque Societ. Reg. Scientiar.
Gall. & Pruss. Sodal.
MATHEMATICI CELEBERRIMI,
ARS CONJECTANDI,
OPUS POSTHUMUM.
Accedit
TRACTATUS
DE SERIEBUS INFINITIS,
Et EPISTOLA Gallicè scripta
DE LUDO PILÆ
RETICULARIS.

Jakob Bernoulli, *Ars Conjectandi*, title page, Basel, 1713, detail. Fondazione Mansutti, Milan. Wikimedia Commons, public domain

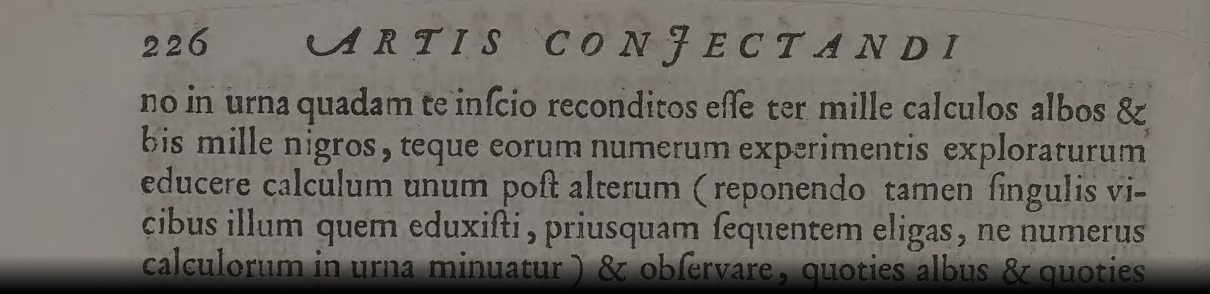

226 ARTIS CONJECTANDI

no in urna quadam te inſcio reconditos eſſe ter mille calculos albos & bis mille nigros, teque eorum numerum experimentis exploraturum educere calculum unum poſt alterum (reponendo tamen ſingulis vicibus illum quem eduxiſti, priusquam ſequentem eligas, ne numerus calculorum in urna minuatur) & obſervare, quoties albus & quoties

"Suppose that, without your knowledge, three thousand white pebbles and two thousand black ones are hidden in an urn. You draw one pebble after another, putting each back, and observe how often a white one and how often a black one comes out."

Jakob Bernoulli, *Ars Conjectandi*, Basel, 1713, part IV, chapter 4. English: lecturer

Jakob Bernoulli, *Ars Conjectandi*, Basel, 1713, page 226, detail. Smithsonian Libraries. Internet Archive, public domain


226 ARTIS CONJECTANDI

no in urna quadam te inſcio reconditos eſſe ter mille calculos albos & bis mille nigros, teque eorum numerum experimentis exploraturum educere calculum unum poſt alterum (reponendo tamen ſingulis vicibus illum quem eduxiſti, priusquam ſequentem eligas, ne numerus calculorum in urna minuatur) & obſervare, quoties albus & quoties

"Suppose that, without your knowledge, three thousand white pebbles and two thousand black ones are hidden in an urn. You draw one pebble after another, putting each back, and observe how often a white one and how often a black one comes out."

After many draws, black comes out about two times in five.
Everyone knows it by instinct; Bernoulli spent twenty years on the proof.

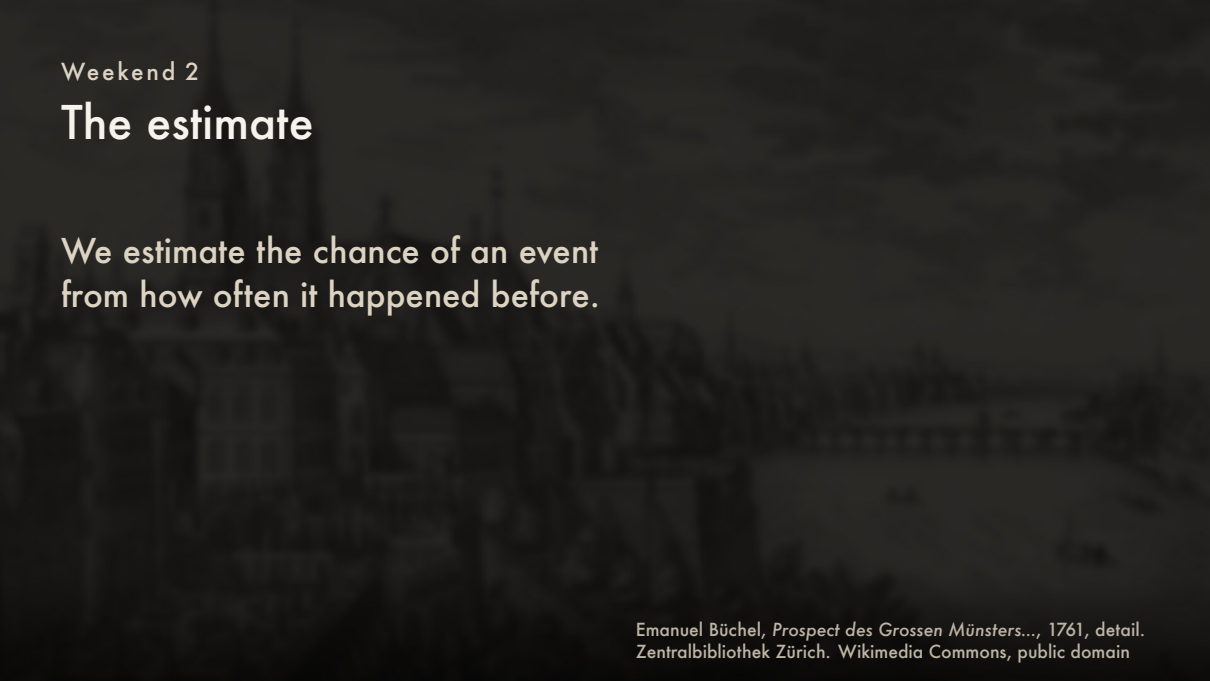
Jakob Bernoulli, *Ars Conjectandi*, Basel, 1713, part IV, chapter 4.
English: lecturer

Jakob Bernoulli, *Ars Conjectandi*, Basel, 1713, page 226, detail.
Smithsonian Libraries. Internet Archive, public domain

Weekend 2

The estimate

We estimate the chance of an event
from how often it happened before.



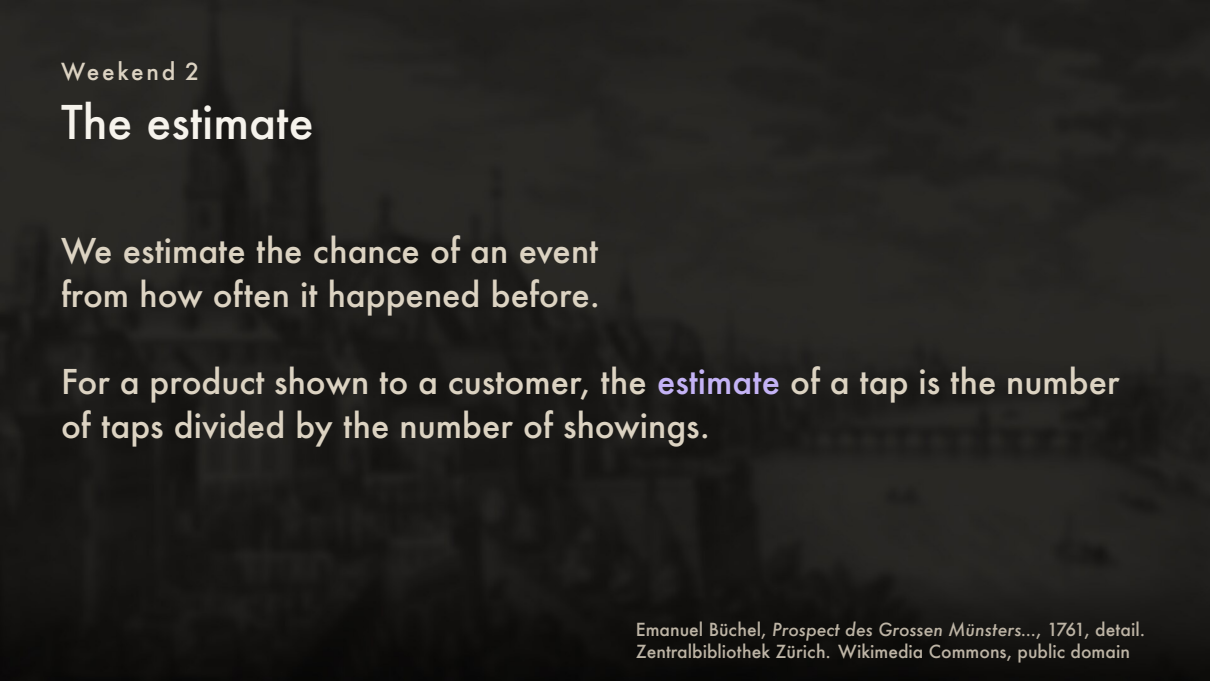
Emanuel Büchel, *Prospect des Grossen Münsters...*, 1761, detail.
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Weekend 2

The estimate

We estimate the chance of an event
from how often it happened before.

For a product shown to a customer, the estimate of a tap is the number of taps divided by the number of showings.



Emanuel Büchel, *Prospect des Grossen Münsters...*, 1761, detail.
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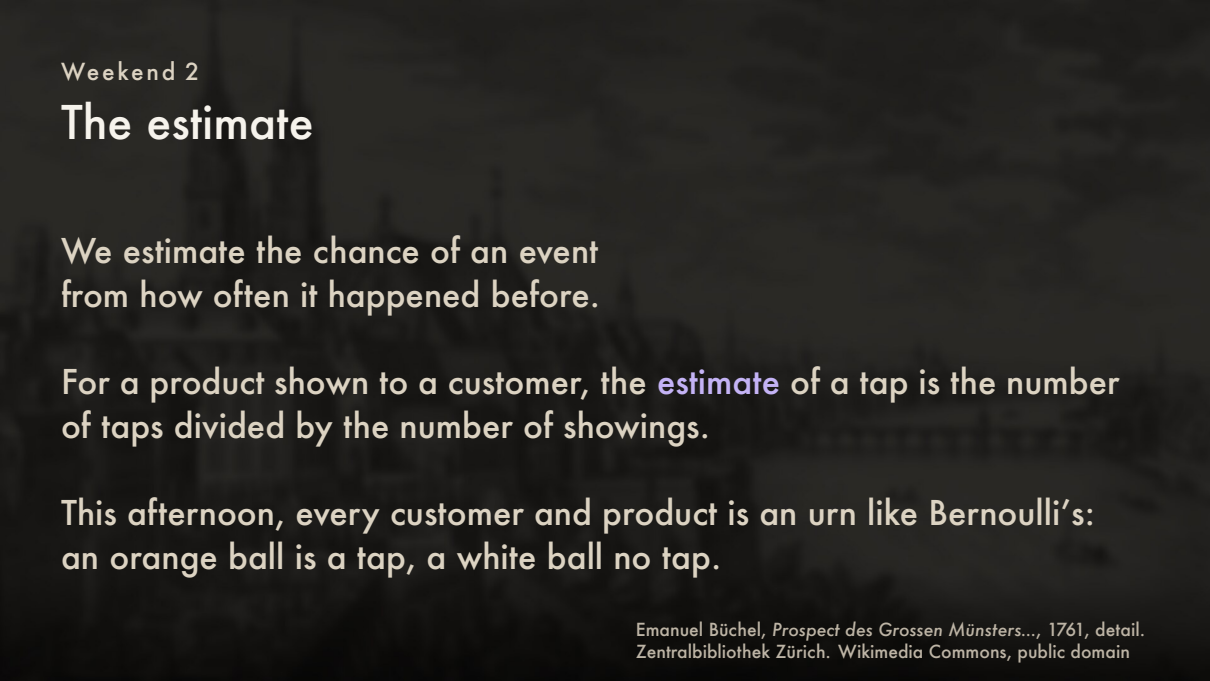
Weekend 2

The estimate

We estimate the chance of an event
from how often it happened before.

For a product shown to a customer, the estimate of a tap is the number of taps divided by the number of showings.

This afternoon, every customer and product is an urn like Bernoulli's: an orange ball is a tap, a white ball no tap.



Emanuel Büchel, *Prospect des Grossen Münsters...*, 1761, detail. Zentralbibliothek Zürich. Wikimedia Commons, public domain

Weekend 2

Estimating chances today

Each option gets an estimate of its chance from the past,
and the most promising is shown.

Spotify mixes of music and podcasts for the home page



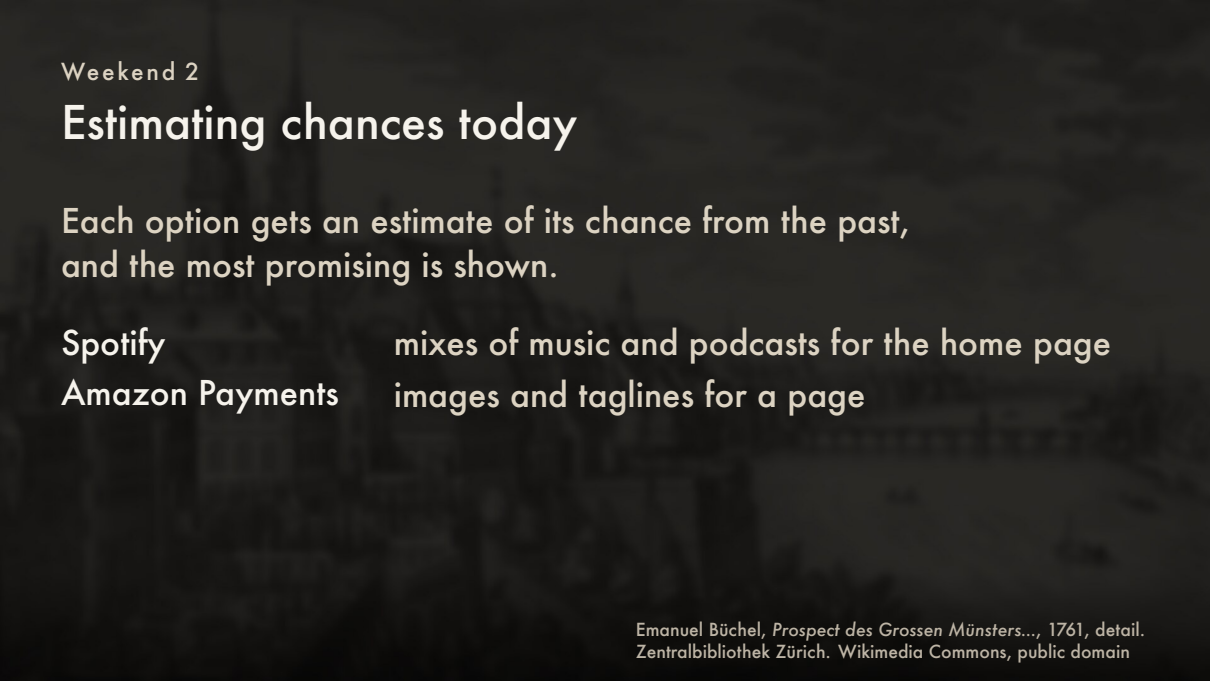
Emanuel Büchel, *Prospect des Grossen Münsters...*, 1761, detail.
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Weekend 2

Estimating chances today

Each option gets an estimate of its chance from the past,
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Spotify	mixes of music and podcasts for the home page
Amazon Payments	images and taglines for a page



Emanuel Büchel, *Prospect des Grossen Münsters...*, 1761, detail.
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Weekend 2

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Spotify	mixes of music and podcasts for the home page
Amazon Payments	images and taglines for a page
Online shops	products for a customer

Emanuel Büchel, *Prospect des Grossen Münsters...*, 1761, detail.
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Emanuel Büchel, *Prospect des Grossen Münsters...*, 1761, detail.
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Weekend 2

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Such estimates decide much of what we are shown online.

Emanuel Büchel, *Prospect des Grossen Münsters...*, 1761, detail.
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